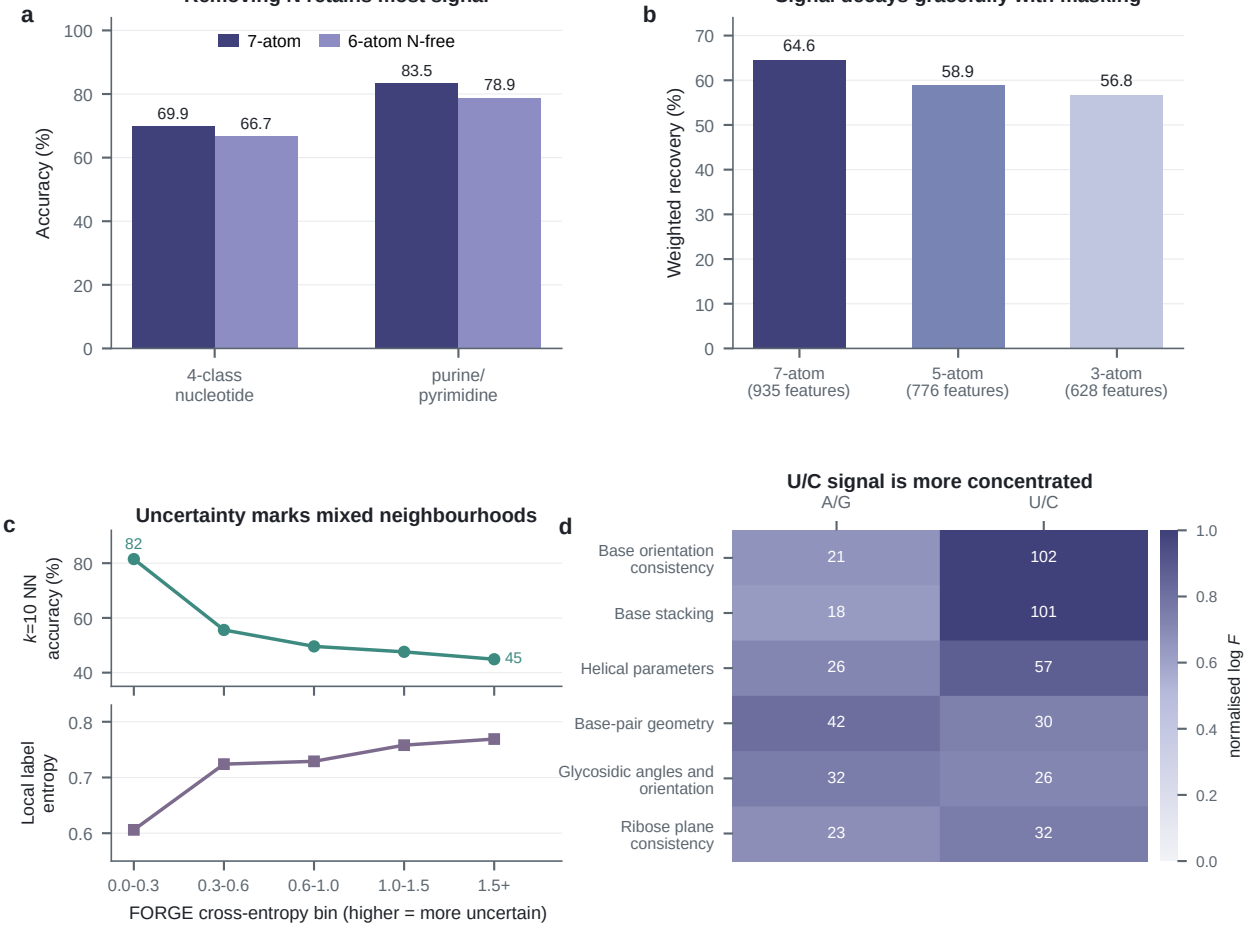
a
Removing N retains most signal
7-atom
6-atom N-free
Accuracy (%)
100
80
60
40
20
0
69.9
66.7
83.5
78.9
4-class nucleotide
purine/ pyrimidine
b
Signal decays gracefully with masking
Weighted recovery (%)
70
60
50
40
30
20
10
0
64.6
58.9
56.8
7-atom (935 features)
5-atom (776 features)
3-atom (628 features)
c
Uncertainty marks mixed neighbourhoods
k=10 NN accuracy (%)
82
45
Local label entropy
0.8
0.7
0.6
0.0-0.3
0.3-0.6
0.6-1.0
1.0-1.5
1.5+
FORGE cross-entropy bin (higher = more uncertain)
d
U/C signal is more concentrated
A/G
U/C
Base orientation consistency
21
102
Base stacking
18
101
Helical parameters
26
57
Base-pair geometry
42
30
Glycosidic angles and orientation
32
26
Ribose plane consistency
23
32
normalised log F
1.0
0.8
0.6
0.4
0.2
0.0